

Python

Logging

```
import logging
import numpy as np

logging.basicConfig(filename='logfile.log',
                    format='%(asctime)s %(levelname)s: %(message)s',
                    filemode='w')

logger = logging.getLogger()
logger.setLevel(logging.DEBUG)
x=np.random.random((2,3,4))
logger.debug(f'x :{x}')
logger.info("Just an information")
logger.warning("Its a warning")
logger.error("Did you try to divide by zero?")
logger.critical("Internet is down")
```

Numpy

np.random.rand(2,3,4) -> standard uniform
 np.random.random((2,3)) -> random_sample from standard uniform
 np.random.uniform(0,5,(2,3)) -> non-standard uniform distribution

np.random.randn(2,3,4) -> standard normal
 np.random.normal(0,5,(2,3)) -> non-standard normal distribution

```
$a1 = np.random.rand(5,10,2)
$b1 = np.random.rand(5,5,2)
$np.hstack((a1,b1)).shape
(5,15,2)
$np.concatenate((a1,b1),axis=1)
a2 = np.random.rand(10,5,2)
b2 = np.random.rand(5,5,2)
$np.vstack((a2,b2)).shape
(15,5,2)
$np.concatenate((a2,b2),axis=0).shape
(15,5,2)
```

np.nonzero(J) -> returns tuples of indices of J that are nonzero
 np.maximum(x1,x2) -> returns the maximum of x1 and x2, element-wise. This is a scalar if both x1 and x2 are scalars.
 np.max(J, axis=1) -> Maximum of a. If axis is None, the result is a scalar value. If axis is an int, the result is an array of dimension `a.ndim - 1`. If axis is a tuple, the result is an array of dimension `a.ndim - len(axis)`.
 np.where(J>t,J,0) -> fills the position where the boolean condition is true with elements of J, 0 otherwise.

Get indices of N maximum values in a numpy array:

```
n=5
arr=np.random.rand(2,3)
x, y = np.unravel_index(np.argsort(arr.ravel(), axis=None)[::-1][:n], arr.shape)
```